

No. of Functional Features Included in the Solution

Date	14 July 2026
Team ID	
Project Name	SMART LENDER
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Predictive Classification	Predicts loan eligibility based on applicant details.	ML Service	Done	2.0
2	Data Preprocessing	Handles missing values and scales numerical inputs.	Data Pipeline	Done	1.0
3	Oversampling Pipeline	Applies SMOTE to handle target class imbalance.	Data Pipeline	Done	0.5
4	Web Interface	Accepts applicant details via interactive HTML forms.	Frontend	Done	0.5
5	Model Comparison	Compares Decision Tree, Random Forest, KNN, and XGBoost.	ML Service	Done	0.5
6	Public Deployment	Tunnels local server using Ngrok for evaluator access.	DevOps	Done	0.5
7					
8					

Feature Summary:

Metric	Count / Value
Total Features Planned	6
Total Features Implemented	6
Core / Must-Have Features	4

Additional / Nice-to-Have Features	2
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Form inputs and result pages	home.html, predict.html, submit.html
2	Backend / Logic	Flask server and prediction API	app.py routes and model load
3	Database / Storage	Applicant historical profiles	loan_data.csv
4	API / Integration	Model prediction endpoints	predict() API
5	Security / Authentication	Validation and safe form input parsing	Input sanitization